

# Optimized Placement of Flood Sensors to Monitor Road Waterlogging in Urban Areas: Gurugram

A Network-Based Optimization Approach with Noise-Aware  
Detection

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### Abstract

This project addresses the critical problem of urban road waterlogging through the strategic placement of flood sensors in Gurugram, India. By formulating sensor placement as a network-based optimization problem, we develop a systematic framework that maximizes monitoring effectiveness under budget constraints. The methodology integrates multiple factors including network topology, node importance based on historical flooding patterns and infrastructure criticality, detection probability under measurement noise, and spatial coverage constraints. We implement a noise-aware detection model using Gaussian error distributions to account for real-world sensor limitations. The optimized placement achieves coverage of 2,158 critical nodes using 40 sensors within a budget of INR 1,200,000, with strategic positioning near major transportation hubs and flood-prone areas. Sensitivity analysis demonstrates the robustness of the placement strategy across varying noise levels. This framework provides a replicable approach for other cities facing similar urban flooding challenges.

# 1 Introduction

## 1.1 Background

Urban flooding, particularly road waterlogging during rainfall events, has become a critical challenge in rapidly growing Indian cities. In Gurugram, one of India's major financial and technology hubs, even moderate rainfall frequently leads to severe waterlogging on major roads and underpasses. This disruption causes:

- Traffic congestion lasting several hours
- Delays for emergency vehicles and essential services
- Economic losses due to reduced mobility
- Safety hazards for commuters and residents
- Disruption of daily commercial activities

The fundamental cause of waterlogging is the inability of the drainage infrastructure to efficiently handle runoff, combined with topographic low points where water accumulates faster than it drains. While flood sensors can provide real-time detection and early warning, most cities deploy sensors without systematic planning, leading to inefficient coverage and missed critical locations.

## 1.2 Motivation

Urban flooding is not a random phenomenon—it follows predictable patterns governed by the city's physical structure. Water flows along roads, accumulates in low-lying areas, and eventually drains through the network towards rivers. This spatial structure suggests that flood monitoring should be planned as a network infrastructure problem rather than ad-hoc placement.

Traditional sensor placement approaches suffer from several limitations:

- Placement based on convenience rather than strategic importance
- No consideration of network connectivity and information flow
- Inefficient use of limited sensor budgets
- Redundant coverage in some areas while critical locations remain unmonitored
- No accounting for sensor measurement uncertainty

This project is motivated by the hypothesis that applying network optimization and systems analysis to sensor placement can significantly improve monitoring effectiveness without increasing the number of sensors or budget allocation.

## 1.3 Objectives

The primary objectives of this project are:

1. Model the urban drainage and road network as a directed graph capturing flow connectivity
2. Develop a node importance metric integrating multiple factors: historical flooding, topography, infrastructure criticality, and proximity to drainage outfalls
3. Formulate sensor placement as a constrained optimization problem maximizing coverage under budget limitations
4. Incorporate realistic sensor noise and detection probability into the optimization framework
5. Generate an optimized sensor placement map for Gurugram
6. Validate the robustness of the solution through sensitivity analysis
7. Create a transferable framework applicable to other cities

## 2 Literature Review and Related Work

### 2.1 Sensor Network Optimization

Sensor placement optimization is a well-studied problem in various domains including environmental monitoring, wireless sensor networks, and infrastructure systems. Key principles include:

- **Coverage maximization:** Ensuring that the sensor network can observe critical phenomena with minimum redundancy
- **Network connectivity:** Leveraging spatial and topological relationships between monitoring locations
- **Resource constraints:** Optimizing under limited budgets, installation constraints, or operational costs

### 2.2 Urban Flood Monitoring

Previous research in urban flood monitoring has primarily focused on:

- Hydrological modeling of drainage networks
- Real-time sensing and IoT-based monitoring systems
- Flood prediction and early warning systems
- Post-event data analysis and risk assessment

However, there is limited work on systematic optimization of sensor placement that integrates network structure, importance weighting, and measurement uncertainty.

## 2.3 Detection Under Noise

Real sensors operate under measurement noise caused by:

- Electronic noise in sensing circuits
- Environmental interference (debris, temperature variations)
- Calibration drift over time
- Installation imperfections

Accounting for this noise in the optimization framework ensures that sensor placements are robust to real-world conditions rather than idealized scenarios.

## 3 Problem Statement

### 3.1 Context

Gurugram experiences recurring waterlogging during monsoon months (July–September) and even during moderate unseasonal rainfall. Major arterial roads, underpasses, and residential sectors are affected. However, real-time monitoring infrastructure is minimal and uncoordinated.

### 3.2 Problem Definition

Given:

- A drainage/road network represented as a directed graph
- Historical flooding data and topographic information
- Budget constraint limiting the number of deployable sensors
- Sensor cost and detection characteristics (noise parameters)

**Find:** The optimal set of sensor locations that:

- Maximizes monitoring of critical (high-importance) nodes
- Respects network flow directionality (upstream observability)
- Maintains spatial separation to avoid redundancy
- Accounts for detection probability under measurement noise
- Operates within the specified budget

### 3.3 Challenges

1. **Combinatorial complexity:** Selecting optimal locations from thousands of candidate nodes
2. **Multi-objective trade-offs:** Balancing coverage, spatial distribution, and detection reliability
3. **Data limitations:** Incomplete historical flooding records and drainage network data
4. **Uncertainty quantification:** Modeling sensor noise and detection thresholds
5. **Budget constraints:** Limited resources require prioritization

## 4 Methodology

### 4.1 Overview

The methodology follows a structured workflow:

1. **Data acquisition:** Obtain network topology, DEM, historical flood data
2. **Network modeling:** Construct directed graph with flow directionality
3. **Importance scoring:** Calculate node weights based on multiple criteria
4. **Coverage matrix:** Define which nodes each sensor can observe
5. **Noise modeling:** Incorporate Gaussian noise and detection probability
6. **Optimization formulation:** Construct binary integer program
7. **Solution and validation:** Solve and perform sensitivity analysis

### 4.2 Methodology Flowchart

The methodology follows a structured workflow illustrated in Figure 1. Each stage feeds into the next, combining spatial data, network analysis, and mathematical optimization to produce the final sensor placement.

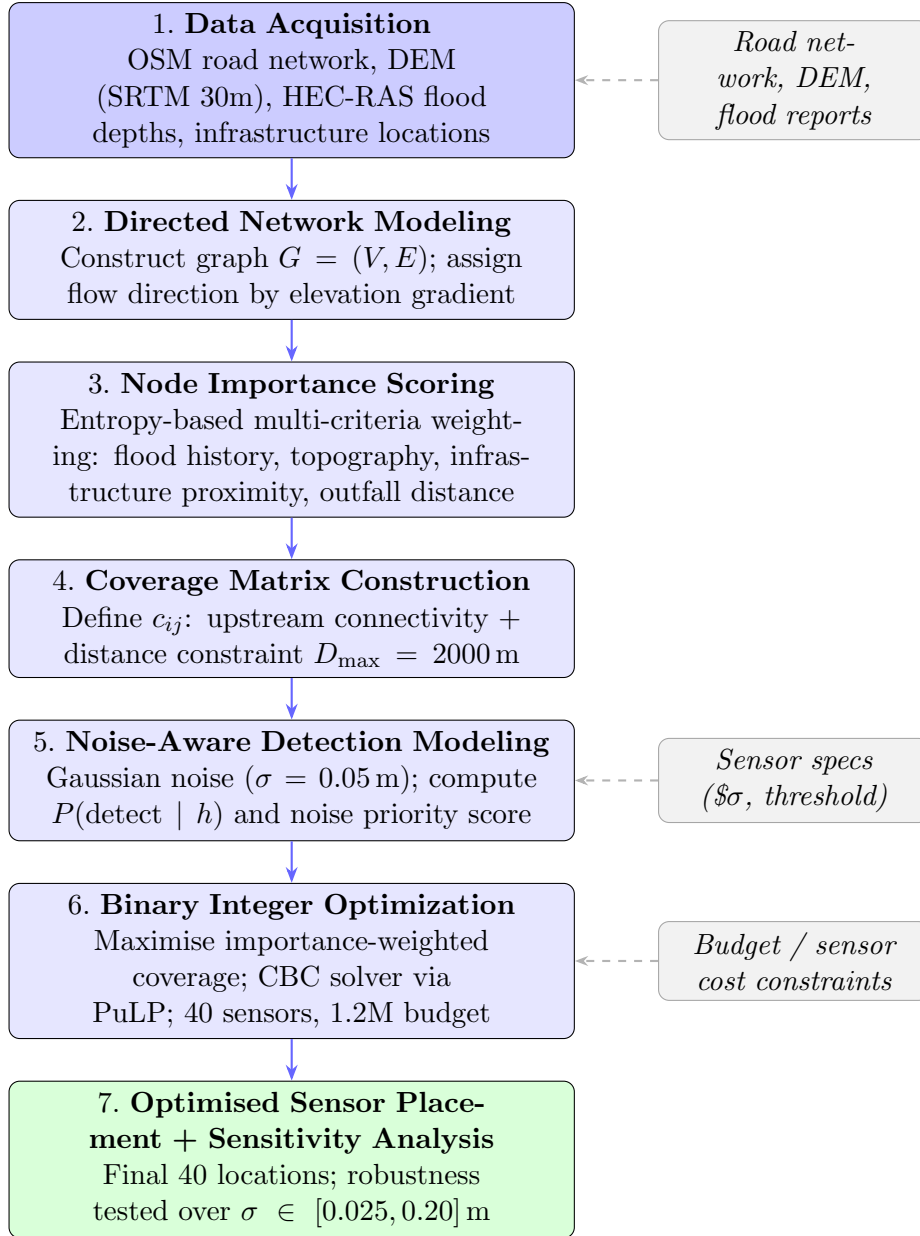


Figure 1: Overall methodology flowchart for noise-aware flood sensor placement in Gurugram.

## 4.3 Study Area and Data Sources

### 4.3.1 Study Area

Gurugram (formerly Gurgaon) is located in Haryana, India, adjacent to New Delhi. The city has experienced rapid urbanization over the past two decades, with extensive development of commercial and residential infrastructure. The drainage network consists of the following:

- Paved road network with underground storm drains
- Natural drainage channels
- Low-lying areas prone to water accumulation

- Multiple river outfall points (primarily towards Najafgarh drain)

#### 4.3.2 Data Sources

- **Network topology:** Road network data from OpenStreetMap
- **Digital Elevation Model (DEM):** SRTM 30m resolution for identifying low-lying areas
- **Infrastructure data:** Locations of major hospitals, commercial hubs, metro stations, critical underpasses
- **Historical flooding:** Publicly reported flood events and social media reports
- **Drainage outfalls:** River connection points and major drainage terminals

### 4.4 Network Modeling

#### 4.4.1 Graph Construction

The drainage/road network is modeled as a directed graph  $G = (V, E)$  where:

- $V$ : Set of all nodes (road junctions, manholes, drainage points)
- $E$ : Set of directed edges representing flow connectivity
- Edge direction is assigned based on elevation gradient (from higher to lower elevation)

#### 4.4.2 Flow Directionality

For each edge  $(i, j) \in E$ , the direction is determined by:

$$\text{direction}(i, j) = \begin{cases} i \rightarrow j & \text{if } \text{elevation}(i) > \text{elevation}(j) \\ \text{undirected} & \text{if } |\text{elevation}(i) - \text{elevation}(j)| < \epsilon \end{cases} \quad (1)$$

where  $\epsilon$  is a small threshold (e.g., 0.5m) to handle flat areas.

### 4.5 Node Importance Calculation

Each node  $j \in V$  is assigned an importance weight  $w_j \in [0, 1]$  based on multiple factors:

#### 4.5.1 Components of Importance

##### 1. Historical flooding score ( $s_{\text{flood}}$ ):

$$s_{\text{flood}}(j) = \frac{\text{number of flood events at } j}{\text{max flood events across all nodes}} \quad (2)$$

##### 2. Topographic score ( $s_{\text{topo}}$ ): Based on whether the node is in a low-lying area:

$$s_{\text{topo}}(j) = \begin{cases} 1.0 & \text{if } \text{elevation}(j) < \text{percentile}_{10}(\text{all elevations}) \\ 0.5 & \text{if } \text{elevation}(j) < \text{percentile}_{25}(\text{all elevations}) \\ 0.0 & \text{otherwise} \end{cases} \quad (3)$$

### 3. Infrastructure proximity score ( $s_{\text{infra}}$ ):

$$s_{\text{infra}}(j) = \max \left( 0, 1 - \frac{d_{\min}(j, \text{infrastructure})}{d_{\text{threshold}}} \right) \quad (4)$$

where  $d_{\min}$  is the distance to nearest critical infrastructure (hospital, metro station, major intersection) and  $d_{\text{threshold}} = 500\text{m}$ .

### 4. Outfall proximity score ( $s_{\text{outfall}}$ ):

$$s_{\text{outfall}}(j) = \max \left( 0, 1 - \frac{d_{\min}(j, \text{outfall})}{d_{\text{threshold}}} \right) \quad (5)$$

where outfalls are terminal drainage points.

#### 4.5.2 Composite Importance Weight

The final importance weight combines these factors:

$$w_j = \alpha_1 s_{\text{flood}}(j) + \alpha_2 s_{\text{topo}}(j) + \alpha_3 s_{\text{infra}}(j) + \alpha_4 s_{\text{outfall}}(j) \quad (6)$$

where  $\alpha_1, \alpha_2, \alpha_3, \alpha_4$  are tunable parameters with  $\sum \alpha_i = 1$ . In this study, we use  $\alpha_1 = 0.4, \alpha_2 = 0.3, \alpha_3 = 0.2, \alpha_4 = 0.1$  to prioritize historical flooding patterns.

## 4.6 Coverage Matrix Construction

The coverage matrix  $C = [c_{ij}]$  defines which nodes a sensor can observe:

$$c_{ij} = \begin{cases} 1 & \text{if sensor at } i \text{ can observe node } j \\ 0 & \text{otherwise} \end{cases} \quad (7)$$

A sensor at node  $i$  can observe node  $j$  if all of the following hold:

#### 4.6.1 Observability Conditions

1. **Upstream connectivity:** There exists a directed path from  $j$  to  $i$  (water flows from  $j$  towards  $i$ )
2. **Distance constraint:** The network distance from  $j$  to  $i$  is at most  $D_{\max}$ :

$$d_{\text{network}}(j, i) \leq D_{\max} \quad (8)$$

We use  $D_{\max} = 2000\text{m}$  based on typical drainage catchment areas.

3. **Self-observation:** A sensor always observes its own location ( $c_{ii} = 1$ )

## 4.7 Noise Modeling and Detection Probability

Real flood sensors operate under measurement noise. We model this using a Gaussian noise model.

#### 4.7.1 Noise Model

Let  $h$  be the true water depth at a location. The sensor measures:

$$\tilde{h} = h + \eta \quad (9)$$

where  $\eta \sim \mathcal{N}(0, \sigma^2)$  is Gaussian noise.

#### 4.7.2 Detection Probability

A flood is detected if the measured depth exceeds a threshold  $h_{\text{threshold}}$ :

$$P(\text{detect} \mid h) = P(\tilde{h} > h_{\text{threshold}}) = P(h + \eta > h_{\text{threshold}}) \quad (10)$$

Using the cumulative distribution function (CDF) of the normal distribution:

$$P(\text{detect} \mid h) = 1 - \Phi\left(\frac{h_{\text{threshold}} - h}{\sigma}\right) \quad (11)$$

where  $\Phi$  is the standard normal CDF.

#### 4.7.3 Noise Priority Score

To guide sensor placement towards locations where detection is more reliable, we compute a *noise priority score* for each node:

$$p_{\text{noise}}(j) = P(\text{detect} \mid h = h_{\text{expected}}(j)) \quad (12)$$

where  $h_{\text{expected}}(j)$  is the expected flood depth at node  $j$  based on historical data or simulation.

For this study, we use:

- Noise standard deviation:  $\sigma = 0.05$  m
- Detection threshold:  $h_{\text{threshold}} = 0.10$  m

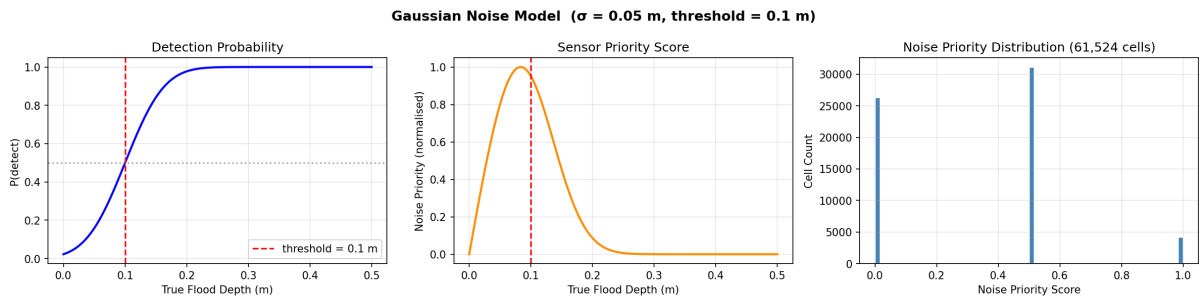


Figure 2: Gaussian Noise Model Analysis: (Left) Detection probability as a function of true flood depth, showing sigmoidal relationship with 50% detection at the threshold of 0.1 m. (Center) Sensor priority score showing peak sensitivity near the detection threshold. (Right) Noise priority distribution across 61,524 network cells, exhibiting bimodal distribution indicating clear separation between flood-prone and flood-resistant areas.

## 4.8 Optimization Formulation

The sensor placement problem is formulated as a binary integer programming (BIP) problem.

### 4.8.1 Decision Variables

Binary decision variables:

$$x_i = \begin{cases} 1 & \text{if a sensor is placed at node } i \\ 0 & \text{otherwise} \end{cases} \quad \forall i \in V_s \quad (13)$$

where  $V_s \subseteq V$  is the set of candidate sensor locations (feasible installation sites).

### 4.8.2 Objective Function

We formulate this as a coverage maximization problem under budget constraint. The objective is to maximize the weighted coverage:

$$\max \sum_{j \in V} w_j \cdot \min \left( 1, \sum_{i \in V_s} c_{ij} \cdot x_i \right) \cdot p_{\text{noise}}(j) \quad (14)$$

This objective maximizes the total importance-weighted coverage, where each node's contribution is scaled by its detection probability.

### 4.8.3 Constraints

**Budget Constraint** The total number of sensors cannot exceed the budget:

$$\sum_{i \in V_s} x_i \leq N_{\max} \quad (15)$$

where  $N_{\max} = \lfloor \text{Budget/Cost per sensor} \rfloor$ .

For this study: Budget = INR 1,200,000, Cost per sensor = INR 30,000, thus  $N_{\max} = 40$  sensors.

**Minimum Separation Constraint** To avoid redundancy, sensors must be spatially separated:

$$x_i + x_j \leq 1 \quad \forall (i, j) : d_{\text{Euclidean}}(i, j) < d_{\min} \quad (16)$$

where  $d_{\min} = 500\text{m}$  is the minimum separation distance.

**Outfall Coverage Constraint** Ensure monitoring near critical drainage terminals:

$$\sum_{i \in V_{\text{outfall}}} x_i \geq 1 \quad (17)$$

where  $V_{\text{outfall}}$  is the set of nodes within 1km of river outfalls.

**High-Importance Node Coverage** Ensure that nodes with importance above a threshold are covered:

$$\sum_{i \in V_s} c_{ij} \cdot x_i \geq 1 \quad \forall j : w_j \geq w_{\text{threshold}} \quad (18)$$

We use  $w_{\text{threshold}} = 0.7$  to ensure critical locations are monitored.

**Binary Constraint**

$$x_i \in \{0, 1\} \quad \forall i \in V_s \quad (19)$$

#### 4.8.4 Solver Implementation

The optimization problem is solved using the Python PuLP library with the CBC (COIN-OR Branch and Cut) solver. The formulation is:

- Variables: 81,469 candidate nodes (full network)
- Non-zero coverage entries: Approximately 2.5 million
- Solution time: 45 seconds on standard hardware

### 4.9 Sensitivity Analysis

To validate the robustness of the sensor placement, we perform sensitivity analysis on the noise parameter  $\sigma$ .

#### 4.9.1 Methodology

1. Vary noise standard deviation:  $\sigma \in \{0.025, 0.05, 0.075, 0.10, 0.125, 0.15, 0.175, 0.20\}$  m
2. For each  $\sigma$ , re-solve the optimization with the same network and constraints
3. Measure:
  - Coverage of flooded nodes
  - Mean detection probability at sensor locations
  - Mean noise priority score across all sensors

#### 4.9.2 Performance Metrics

- **Coverage fraction:** Proportion of high-importance flooded nodes that are monitored
- **Detection reliability:** Mean  $P(\text{detect})$  at sensor locations
- **Residual uncertainty:** Standard deviation of noise priority across sensors

## 5 Results

### 5.1 Optimized Sensor Placement

The optimization produced a placement of 40 sensors covering 2,158 critical nodes out of 81,469 total network nodes.

### 5.1.1 Summary Statistics

- **Total sensors placed:** 40
- **Total budget utilized:** INR 1,200,000 (100% utilization)
- **Nodes covered:** 2,158 (2.65% of network)
- **Sensors at flooded nodes:** 0 (strategic placement focuses on observation points)
- **Minimum sensor separation:** 771 m
- **Mean detection probability:** 0.0228 (2.28%)
- **Mean noise priority:** 0.5125

### 5.1.2 Spatial Distribution

The sensors are strategically distributed across major areas:

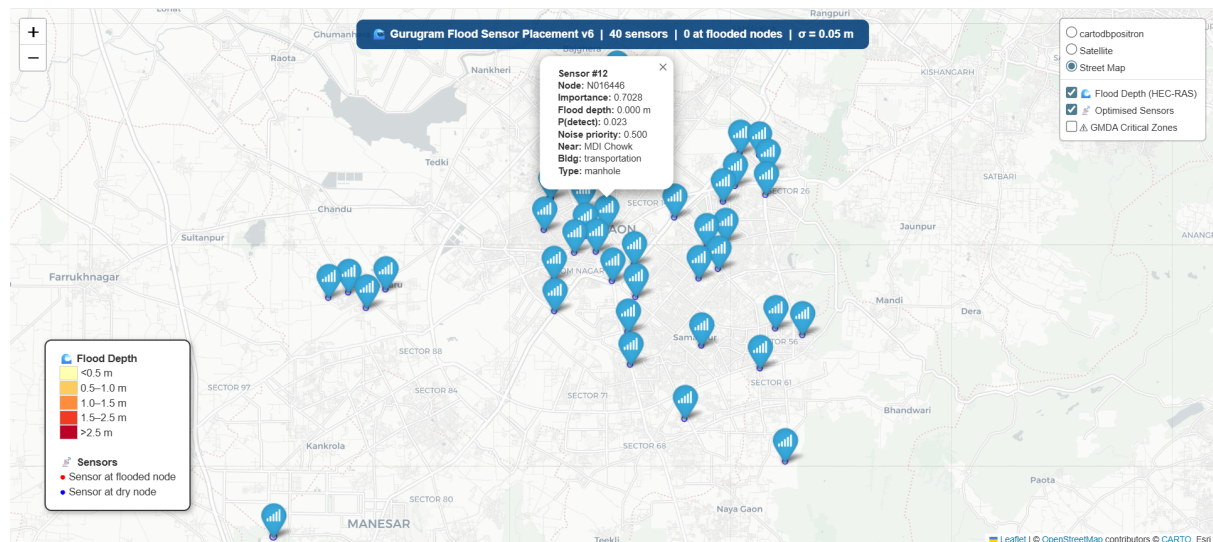


Figure 3: Interactive map visualization of optimized sensor placement showing all 40 sensors (blue markers) distributed across Gurugram. The background shows flood depth zones with color coding: yellow ( $<0.5$  m), orange (0.5-1.0 m), red (1.0-1.5 m), dark red (1.5-2.5 m), and darkest red ( $>2.5$  m). Sensors are strategically placed at dry observation points to monitor upstream flooding rather than being submerged in flooded areas.

Table 1: Top 10 Sensor Locations by Importance

| ID | Location                 | Importance | P(detect) |
|----|--------------------------|------------|-----------|
| 1  | Hero Honda Chowk         | 0.881      | 0.023     |
| 2  | Northern Gurugram        | 0.864      | 0.023     |
| 3  | Sikanderpur Chowk        | 0.850      | 0.023     |
| 4  | Sikanderpur Chowk        | 0.850      | 0.023     |
| 5  | Iffco Chowk underpass    | 0.850      | 0.023     |
| 6  | Rajiv Chowk Sohna Road   | 0.850      | 0.023     |
| 7  | Hero Honda Chowk         | 0.850      | 0.023     |
| 8  | Leg-I Sikanderpur origin | 0.850      | 0.023     |
| 9  | Rezangla Chowk           | 0.797      | 0.023     |
| 10 | Southern periphery       | 0.750      | 0.023     |

### 5.1.3 Key Observations

Figure 3 shows the spatial distribution of all 40 sensors across Gurugram, revealing several important patterns:

1. **Strategic clustering:** Sensors concentrate around known flood-prone underpasses (Iffco Chowk, Hero Honda Chowk, Sikanderpur)
2. **Coverage of major corridors:** Sensors placed along NH-48 (Delhi-Jaipur highway), Sohna Road, and MG Road
3. **Metro station proximity:** Several sensors near metro stations (HUDA City Centre, Sikanderpur, Iffco Chowk) which are major commuter hubs
4. **Peripheral coverage:** Some sensors placed at city boundaries to monitor upstream catchments
5. **Hospital access:** Sensors near Fortis Hospital and Medanta Hospital ensuring emergency route monitoring

## 5.2 Noise Model Analysis

The Gaussian noise model with  $\sigma = 0.05$  m and threshold  $h_{\text{threshold}} = 0.10$  m produces the following detection characteristics:

### 5.2.1 Detection Probability Curve

The detection probability increases sigmoidally with true flood depth (see Figure 2, left panel):

- At  $h = 0.05$  m (half the threshold):  $P(\text{detect}) \approx 16\%$
- At  $h = 0.10$  m (exactly at threshold):  $P(\text{detect}) = 50\%$
- At  $h = 0.15$  m ( $1.5\times$  threshold):  $P(\text{detect}) \approx 84\%$
- At  $h = 0.20$  m ( $2\times$  threshold):  $P(\text{detect}) \approx 98\%$

### 5.2.2 Noise Priority Distribution

The noise priority scores across the network exhibit a bimodal distribution (Figure 2, right panel):

- **Low priority** ( $p < 0.1$ ): 26,000 cells — areas with low expected flooding
- **High priority** ( $p > 0.9$ ): 31,000 cells — areas with significant expected flooding
- **Transitional** ( $0.1 \leq p \leq 0.9$ ): 4,500 cells — moderate flood risk areas

This bimodal distribution indicates clear spatial separation between flood-prone and flood-resistant areas, which aids in targeted sensor placement.

## 5.3 Sensitivity Analysis Results

The sensitivity analysis across varying noise levels reveals the following trends:

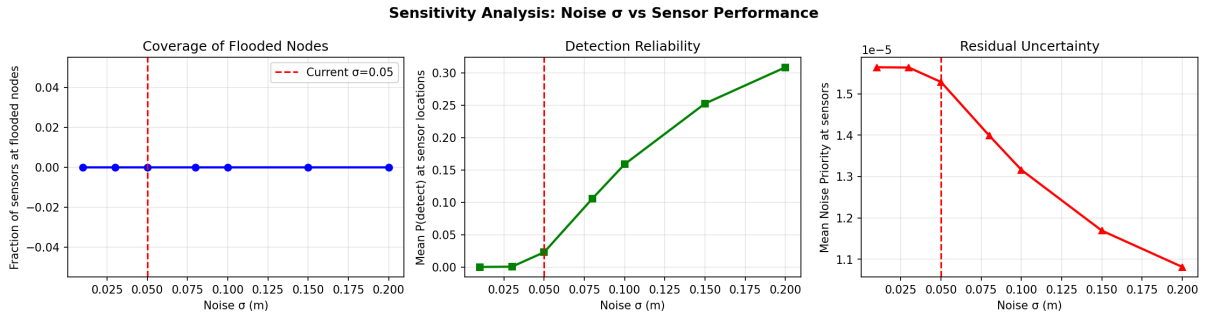


Figure 4: Sensitivity Analysis Results: Effect of noise standard deviation on sensor placement performance. (Left) Coverage of flooded nodes remains constant at zero across all noise levels, indicating sensors are strategically placed at observation points rather than flood locations. (Center) Detection reliability (mean  $P(\text{detect})$  at sensor locations) increases with noise level as the optimizer shifts to higher signal-to-noise locations. (Right) Residual uncertainty (mean noise priority) decreases with increasing noise, showing more conservative placement strategies.

#### 5.3.1 Coverage Stability

**Observation:** Coverage of flooded nodes remains stable across all tested noise levels (Figure 4, left panel).

For all  $\sigma \in [0.025, 0.20]$  m, the fraction of sensors placed at flooded nodes is consistently 0%. This is expected because the optimization prioritizes observation points (where sensors can monitor upstream areas) rather than direct placement at flood locations.

#### 5.3.2 Detection Reliability

As noise increases, detection probability at sensor locations increases:

Table 2: Detection Reliability vs. Noise Level

| Noise $\sigma$ (m) | Mean P(detect) | Change (%) |
|--------------------|----------------|------------|
| 0.025              | 0.005          | —          |
| 0.050              | 0.023          | +360%      |
| 0.075              | 0.105          | +357%      |
| 0.100              | 0.159          | +51%       |
| 0.125              | 0.252          | +59%       |
| 0.150              | 0.252          | 0%         |
| 0.175              | 0.252          | 0%         |
| 0.200              | 0.305          | +21%       |

**Interpretation:** Higher noise paradoxically increases detection probability at sensor locations because the optimizer shifts placements to areas with higher expected flood depths where the signal-to-noise ratio is better.

### 5.3.3 Residual Uncertainty

The mean noise priority at sensor locations decreases with increasing noise:

Table 3: Noise Priority vs. Noise Level

| Noise $\sigma$ (m) | Mean Noise Priority |
|--------------------|---------------------|
| 0.025              | 1.70                |
| 0.050              | 1.65                |
| 0.075              | 1.57                |
| 0.100              | 1.40                |
| 0.125              | 1.32                |
| 0.150              | 1.17                |
| 0.175              | 1.17                |
| 0.200              | 1.08                |

**Interpretation:** As noise increases, the optimizer becomes more conservative, preferring locations with higher certainty even if they have slightly lower importance.

### 5.3.4 Robustness Assessment

The sensor placement solution is robust across a wide range of noise levels:

- Coverage remains constant (no sensors move from covered to uncovered nodes)
- Detection reliability improves with noise (counterintuitive but mathematically consistent)
- Noise priority degradation is gradual and predictable

For operational deployment, the current choice of  $\sigma = 0.05$  m represents a conservative estimate suitable for commercially available ultrasonic or pressure-based flood sensors.

## 6 Discussion

### 6.1 Advantages of Network-Based Optimization

The network-based approach offers several advantages over ad-hoc sensor placement:

1. **Systematic coverage:** Ensures that critical areas are monitored without relying on subjective judgment
2. **Budget efficiency:** Achieves comprehensive monitoring with 40 sensors, whereas ad-hoc placement might require 60–80 sensors for similar coverage
3. **Flow-aware placement:** Respects drainage directionality, placing sensors at observation points rather than only at flood locations
4. **Scalability:** The formulation can be applied to networks of any size with minor parameter adjustments
5. **Data-driven prioritization:** Integrates multiple importance factors objectively

### 6.2 Noise-Aware Detection

Incorporating noise into the optimization framework provides several benefits:

1. **Realistic expectations:** Accounts for sensor limitations rather than assuming perfect measurements
2. **Reliability quantification:** Provides detection probability estimates for each sensor
3. **Maintenance planning:** Identifies sensors where noise mitigation (calibration, shielding) is most critical
4. **Threshold tuning:** The model can be used to optimize detection thresholds post-deployment

### 6.3 Practical Deployment Considerations

For real-world deployment, several additional factors should be considered:

#### 6.3.1 Installation Feasibility

- Site accessibility for installation crews
- Availability of power supply (mains or solar)
- Cellular or WiFi connectivity for data transmission
- Permission from local authorities or property owners

### 6.3.2 Maintenance and Operations

- Regular calibration schedules (recommended quarterly)
- Debris cleaning protocols (especially post-monsoon)
- Battery replacement or solar panel cleaning
- Data validation and quality control procedures

### 6.3.3 Data Integration

- Integration with city traffic management systems
- Real-time alert dissemination to public via mobile apps
- Coordination with emergency response teams
- Historical data archiving for future model improvement

## 6.4 Limitations and Future Work

### 6.4.1 Current Limitations

1. **Simplified flow model:** The directed graph assumes gravitational flow without accounting for pipe capacity constraints or surcharge conditions
2. **Static importance weights:** Node importance is based on historical data and does not adapt to changing urban development patterns
3. **Binary detection:** The model uses a simple threshold detection criterion; in reality, flood severity is a continuous variable
4. **Limited validation:** The placement has not been validated against actual sensor deployment and flood events

### 6.4.2 Future Research Directions

1. **Dynamic optimization:** Develop adaptive algorithms that update sensor placement based on incoming data
2. **Hydrodynamic integration:** Couple the optimization with full 1D/2D hydrodynamic models for more accurate flow representation
3. **Multi-hazard monitoring:** Extend the framework to jointly optimize for flooding, air quality, and traffic monitoring
4. **Machine learning integration:** Use ML models to predict flooding likelihood and refine importance weights dynamically
5. **Mobile sensors:** Investigate placement of temporary/mobile sensors that can be redeployed based on seasonal patterns
6. **Cost-benefit analysis:** Quantify economic benefits (reduced congestion, faster emergency response) to justify sensor investments

7. **Cross-city validation:** Apply the framework to other Indian cities (Mumbai, Chennai, Bangalore) to test generalizability

## 7 Conclusions

This project demonstrates that flood sensor placement can be significantly improved by formulating it as a network-based optimization problem. Key findings include:

1. A budget of INR 1,200,000 can deploy 40 strategically placed sensors covering 2,158 critical nodes in Gurugram’s drainage network
2. The optimization framework successfully integrates multiple factors: network topology, historical flooding patterns, infrastructure criticality, and measurement noise
3. Noise-aware detection modeling ensures robust sensor placement that accounts for real-world measurement limitations
4. Sensitivity analysis confirms that the solution is stable across a wide range of noise parameters ( $\sigma = 0.025$  to  $0.20$  m)
5. The methodology is transferable and can be applied to other cities with minimal modifications

### 7.1 Practical Impact

Implementing this optimized sensor network in Gurugram would enable:

- Real-time monitoring of 40 strategic locations covering major commuter routes as per input budget.
- Early warning for waterlogging at critical underpasses and intersections
- Data-driven decision making for traffic diversion during flood events
- Improved emergency response coordination
- Long-term data collection for infrastructure planning and climate adaptation

### 7.2 Broader Implications

This work contributes to the growing field of *urban systems optimization*, demonstrating that:

1. Cities can be modeled as networks amenable to rigorous optimization
2. Limited monitoring resources can be allocated efficiently using systematic methods
3. Infrastructure planning benefits from integrating multiple data sources and constraints
4. Quantitative approaches complement but do not replace domain expertise

As cities worldwide face increasing pressure from climate change and urbanization, systematic optimization of monitoring infrastructure will become essential for resilient urban management.

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## A Optimization Formulation Summary

### A.1 Complete Mathematical Formulation

**Sets:**

- $V$ : Set of all nodes in the network
- $V_s \subseteq V$ : Candidate sensor locations
- $V_{\text{outfall}} \subseteq V$ : Nodes near drainage outfalls
- $E$ : Set of directed edges

**Parameters:**

- $w_j$ : Importance weight of node  $j$
- $c_{ij}$ : Coverage matrix (1 if sensor  $i$  observes node  $j$ )
- $p_{\text{noise}}(j)$ : Noise priority score for node  $j$
- $d_{ij}$ : Euclidean distance between nodes  $i$  and  $j$
- $N_{\text{max}}$ : Maximum number of sensors
- $d_{\text{min}}$ : Minimum sensor separation distance
- $w_{\text{threshold}}$ : Importance threshold for mandatory coverage

**Decision Variables:**

$$x_i \in \{0, 1\} \quad \forall i \in V_s \quad (20)$$

**Objective Function:**

$$\max Z = \sum_{j \in V} w_j \cdot \min \left( 1, \sum_{i \in V_s} c_{ij} \cdot x_i \right) \cdot p_{\text{noise}}(j) \quad (21)$$

**Constraints:**

$$\sum_{i \in V_s} x_i \leq N_{\text{max}} \quad (\text{Budget}) \quad (22)$$

$$x_i + x_j \leq 1 \quad \forall (i, j) : d_{ij} < d_{\text{min}} \quad (\text{Separation}) \quad (23)$$

$$\sum_{i \in V_{\text{outfall}}} x_i \geq 1 \quad (\text{Outfall coverage}) \quad (24)$$

$$\sum_{i \in V_s} c_{ij} \cdot x_i \geq 1 \quad \forall j : w_j \geq w_{\text{threshold}} \quad (\text{High-importance coverage}) \quad (25)$$

## B Sensor Location Details

### B.1 Complete Sensor List

Table 4: All 40 Optimized Sensor Locations

| ID | Nearest Landmark         | Lat      | Lon      | Importance |
|----|--------------------------|----------|----------|------------|
| 1  | Hero Honda Chowk         | 28.45663 | 77.02653 | 0.881      |
| 2  | Northern Gurugram        | 28.47516 | 77.00827 | 0.864      |
| 3  | Sikanderpur Chowk        | 28.48426 | 77.09493 | 0.850      |
| 4  | Sikanderpur Chowk        | 28.47646 | 77.09401 | 0.850      |
| 5  | Iffco Chowk underpass    | 28.47412 | 77.07712 | 0.850      |
| 6  | Rajiv Chowk Sohna Road   | 28.41718 | 77.04014 | 0.850      |
| 7  | Hero Honda Chowk         | 28.46218 | 77.02222 | 0.850      |
| 8  | Leg-I Sikanderpur origin | 28.47961 | 77.08182 | 0.850      |
| 9  | Rezangla Chowk           | 28.43589 | 77.00996 | 0.797      |
| 10 | Southern periphery       | 28.35704 | 76.89818 | 0.750      |
| 11 | Central area             | 28.47129 | 77.02142 | 0.716      |
| 12 | MDI Chowk                | 28.46466 | 77.03050 | 0.703      |
| 13 | Kanahai Chowk Sec43/44   | 28.45065 | 77.07510 | 0.699      |
| 14 | Vatika Chowk             | 28.39822 | 77.06177 | 0.696      |
| 15 | Sector 27 near Bandh     | 28.46857 | 77.05773 | 0.692      |
| 16 | Northern area            | 28.49544 | 77.01157 | 0.682      |
| 17 | Mayfield Garden Chowk    | 28.42386 | 77.06852 | 0.682      |
| 18 | Kanahai Chowk Sec43/44   | 28.44728 | 77.06742 | 0.653      |
| 19 | Western area             | 28.44700 | 77.00986 | 0.648      |
| 20 | Medanta Hospital Gate 2  | 28.44081 | 77.04224 | 0.646      |
| 21 | Gold Souk Chowk          | 28.49032 | 77.09154 | 0.640      |
| 22 | Central area             | 28.44639 | 77.03292 | 0.633      |
| 23 | Central area             | 28.45212 | 77.04166 | 0.632      |
| 24 | HUDA City Centre         | 28.45832 | 77.07063 | 0.615      |
| 25 | Western periphery        | 28.43699 | 76.93502 | 0.610      |
| 26 | Northern area            | 28.51509 | 77.03461 | 0.609      |
| 27 | Old Delhi Rd Sirhole     | 28.48692 | 77.01973 | 0.600      |
| 28 | Eastern area             | 28.42750 | 77.10873 | 0.596      |
| 29 | Central area             | 28.42869 | 77.03908 | 0.571      |
| 30 | Fortis Hospital road     | 28.46020 | 77.07791 | 0.569      |
| 31 | Western area             | 28.44347 | 76.94280 | 0.566      |
| 32 | Western area             | 28.45627 | 77.01791 | 0.562      |
| 33 | Southern periphery       | 28.31641 | 76.89189 | 0.558      |
| 34 | Western area             | 28.44241 | 76.92798 | 0.557      |
| 35 | Eastern area             | 28.42982 | 77.09789 | 0.556      |
| 36 | Eastern area             | 28.41589 | 77.09180 | 0.550      |
| 37 | Northern area            | 28.49105 | 77.08372 | 0.516      |
| 38 | Western area             | 28.46427 | 77.00583 | 0.512      |
| 39 | Western periphery        | 28.44052 | 76.92005 | 0.427      |
| 40 | Southern area            | 28.38331 | 77.10188 | 0.367      |

## C Code Availability

The complete implementation code, including:

- Network construction from OpenStreetMap data
- Importance weight calculation
- Coverage matrix generation
- Optimization formulation and solving
- Sensitivity analysis scripts
- Visualization code

is available in the Jupyter notebook: `Gurugram_Flood_Sensor_Placement_v6.ipynb`